

Slow light brings faster communications

Two teams of researchers have managed to significantly reduce the speed of light in an optical fibre, which could open the door to all-optical routers for telecommunications, as **Daniel Gauthier** explains

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In the last 10 years, technology that can encode information onto pulses of light and then send it around the globe through ultra-thin glass fibres has revolutionized the way we communicate. With the pulses travelling at speeds of up to $200\,000\text{ km s}^{-1}$, and suffering very little attenuation, someone in New York can have a telephone conversation with someone in Bangalore almost in real time, even though they are some $13\,300\text{ km}$ apart. This is an excellent example of technology bringing people closer together and leading to a truly global economy.

Current telecommunications networks and equipment have little trouble transmitting thousands of phone calls simultaneously with very small delays because telephone conversations actually contain relatively little information. However, if you try to ratchet up the amount of information that you want to transmit, such as sending high-definition video signals, the system soon slows down.

Optical engineers around the globe are working hard to meet the ever-growing demand for higher-speed information networks, and the latest systems being developed operate at rates close to 160 GB per second – which is over 100 times quicker than the fastest broadband services currently available and a world away from the 56 kB per second dial-up connections of the early years of the Internet. Paradoxically, it seems that making light travel slower rather than faster might be the best way to meet these high-speed challenges.

Optical routers

The backbone of the global communication system is the web of optical fibres that criss-crosses the planet. Equally important, however, are devices called routers that switch pulses of light from one fibre to another so that they end up at the correct destination. Currently, routers must first convert the optical pulses into electrical ones, perform the switching operation, and then convert the electrical signal back into an optical pulse. However, this process can create bottlenecks, and it is generally believed that all-optical routers are the best way to achieve the highest data rates.

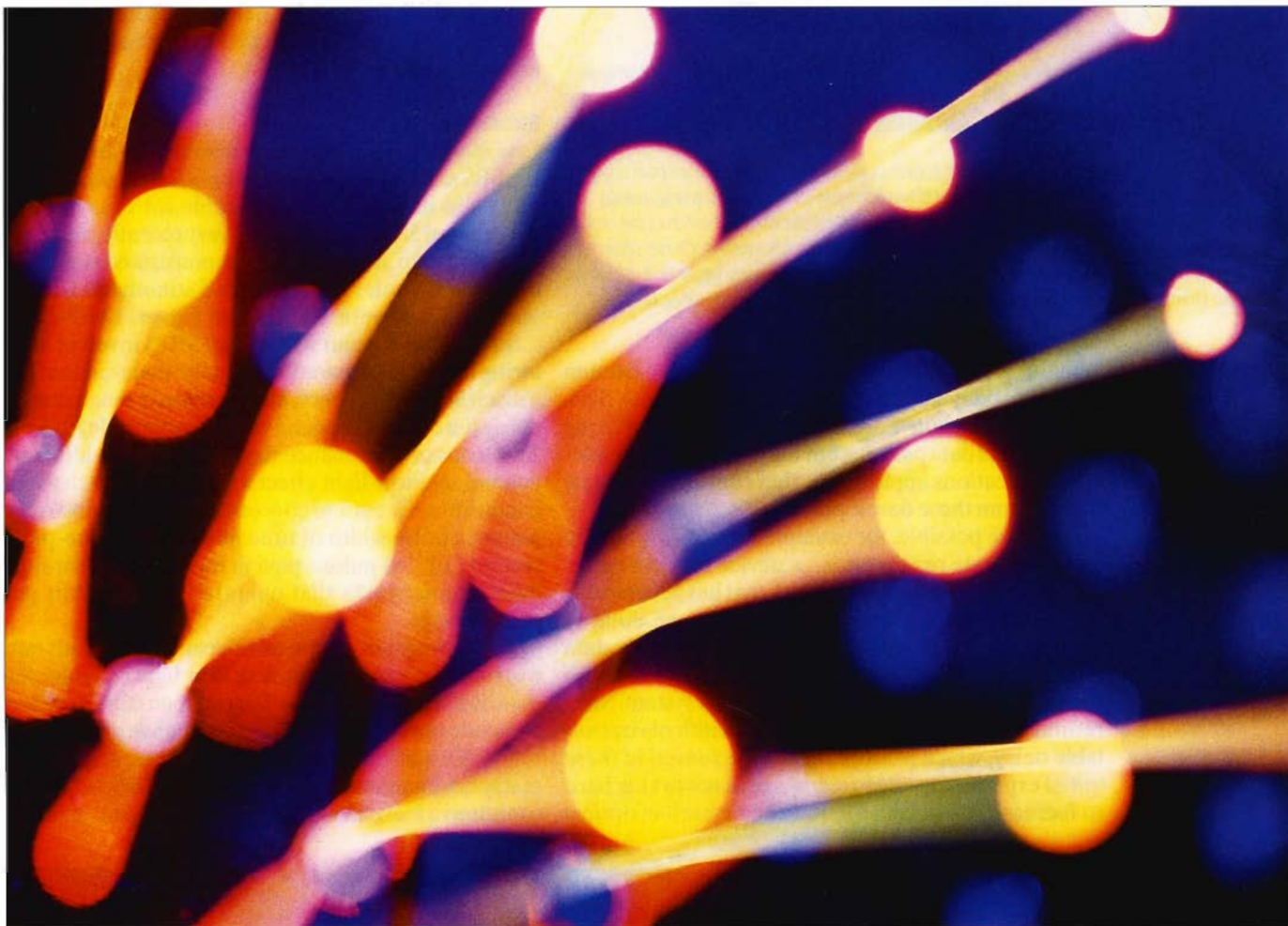
One of the main difficulties in designing an all-optical router is to find a way to temporarily store or “buffer” the packets of information – a feat that is currently performed electronically using high-speed memory chips. Packets are the basic units of data for communications that rely on the internet protocol (IP), with each packet

containing between 10 and $10\,000$ individual pulses of light. However, routers can only process a single packet at a time, so the information sometimes needs to be buffered to prevent packets from different conversations getting mixed up. An ideal buffer would be able to adjust the timing of individual packets to optimize the flow of information through the router.

One possible way to directly buffer an optical signal is to reduce the speed of light itself. In 1992 Steve Harris and co-workers at Stanford University demonstrated that “slow light” could be produced by bombarding a gas of cold atoms with a laser. This “pump” laser modifies the electronic structure of the atoms and causes the refractive index of the gas – as experienced by a weaker “probe” laser – to vary rapidly as a function of the frequency of the light. This, in turn, reduces the group velocity – the speed at which the peak of a laser pulse moves – of the probe laser. Indeed, a similar method can be used to make the group velocity faster than the speed of light in a vacuum, which is also useful when making an optical buffer, and can even bring light to a complete standstill (see *Physics World* October 2004 pp31–34).

Techniques for making slow light advanced rapidly in the 1990s, and by 1998 Lene Hau and co-workers at Harvard University and Stanford had managed to reduce the group velocity of a light pulse to about 17 m s^{-1} – which is comparable to the speed of a racing cyclist (see *Physics World* September 2001 p35–40). Unfortunately, the effect only works when the difference between the “carrier” frequency of the pump and probe lasers precisely matches an atomic resonance and for pulses that are several microseconds in duration, which restricts its use to very low data rates. Moreover, the experimental apparatus is complex and not well suited to real-world applications.

Paradoxically, it seems that making light travel slower rather than faster might be the best way to meet the high-speed challenge



Off the shelf

Two groups have now independently demonstrated that light can be delayed in a controlled manner by exploiting a process called stimulated Brillouin scattering in standard optical fibres. Miguel González-Herráez, currently working in Luc Thévenaz's group at the Ecole Polytechnique Fédérale de Lausanne, Switzerland, and co-workers managed to slow the group velocity of light by a factor of three using off-the-shelf equipment under normal conditions (*Appl. Phys. Lett.* **87** 081113). Meanwhile, Yoshitomo Okawachi, Alexander Gaeta and colleagues at Cornell University, along with co-workers at the University of Rochester and Duke University, including the present author, have used the same technique to delay pulses as short as 15 ns (*Phys. Rev. Lett.* **94** 153902).

In stimulated Brillouin scattering (SBS) the data pulses enter the optical fibre from one end while a control beam enters from the opposite end (see figure over). If the carrier frequency of the data pulses is slightly lower than the frequency of the control beam, the refractive index of the fibre varies rapidly and the pulses can be amplified and/or delayed, depending on the experimental conditions.

The refractive index changes because a process known as electrostriction (the tendency of materials to become compressed in the presence of an electric field) produces density fluctuations or sound waves in the fibre. When the frequency difference between the data pulses and the control beam is equal to a certain value

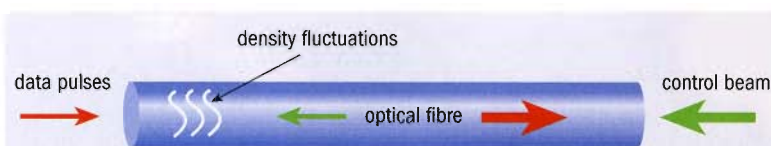
called the Brillouin frequency shift, which is about 11 GHz in standard optical fibres, the two beams interfere to create sound waves with this frequency. When the control beam scatters off these sound waves, its frequency is therefore lowered such that it is equal to the carrier frequency of the data pulses. As a result, the scattered light adds constructively to the data pulses and amplifies them.

This scattering process only occurs over a narrow range of frequencies, which means that the control beam creates a resonant region in which the response of the fibre to light is maximal. This resonance has a width of approximately 30 MHz in a standard telecommunications optical fibre. However, a big advantage of this approach is that the central frequency of the resonance that is responsible for the slow-light effect can be changed by simply changing the frequency of the control beam. To achieve such flexibility in an atomic gas you would have to search around to find an atom or molecule that had a resonance precisely at the desired wavelength, which is virtually impossible.

Both teams have demonstrated that the slow-light effect works at all the frequencies currently used for telecommunications for pulses with durations of about 20 ns, although neither team has yet observed the very low group velocities observed in atomic gases. Fortunately, however, it is not the size of the group velocity that matters: it is much more important to be able to control the length of time by which the pulse is delayed relative to the duration of the pulse.

Take your time

Optical fibres could improve communication networks by reducing the speed of laser pulses.



Slowing light In stimulated Brillouin scattering, data pulses enter the optical fibre from the left (red) and a control or "pump" beam enters from the right (green). The beams create density fluctuations or sound waves in the fibre via a process called electrostriction, which cause some of the light from the control beam to scatter in the direction of the data pulses. Constructive interference between the scattered and incident light amplifies the pulses, which, in turn, leads to a rapid change in the refractive index of the fibre and hence a slower group velocity.

In this regard, both teams have observed maximum delays of about a single pulse width (about 25 ns). Such small relative delays are not yet useful for buffering large optical data packets, but there may be other telecommunications applications that could immediately benefit from these developments.

It may be possible, for example, to use slow light in an optical data regenerator – a device that can recreate a stream of optical data pulses that has become distorted after propagating a very long distance along an optical fibre. It is important to ensure that the data stream is shifted in time so that it is synchronized with the master clock of the communications system. Such retiming requires only a single pulse width of controllable delay, which has already been achieved in these initial experiments. Both teams are now working hard to increase this delay so that it can be used in optical

buffers, which will require delays of about 1000 pulse widths in order to store the data packets processed by modern routers.

Small world

One shortcoming of these experimental advances is that the SBS slow-light effect only works for pulses of about 10 ns and hence is only applicable for data rates of the order of 50 MB per second. Very recently, however, Gaeta's group at Cornell has demonstrated slow-light delay via a similar process called stimulated Raman scattering in an optical fibre (*Opt. Express* **13** 6092).

Stimulated Raman scattering (SRS) involves scattering light from a sound wave in the material that has a much higher frequency, and therefore a much shorter decay time, than the sound waves responsible for SBS. This leads to a resonance that has a larger width, and enables the slow-light effect to work over much wider bandwidths. Gaeta and co-workers have demonstrated almost a pulse width of tuneable delay for sub-pico-second (10^{-12} s) pulses, paving the way for ultra-high data-rate networks that operate in the terabits per second range.

The throughput of optical telecommunications networks will be vastly improved if slow light can solve the optical-buffering problem. We can then start to dream about the possibility of real-time exchange of high-definition video across the globe, or even complete virtual-reality environments. It seems that the world will indeed continue to feel like it is getting smaller all the time.

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Image: Colour coded x-y displacements caused by rotation of a speckle pattern around its centre P Somers and N Bhattacharya 2005 Journal of Optics A: Pure and Applied Optics **7** S385–S391

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